# Taha Berk Koç

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## Education

**Bogazici University** Sept. 2021 – May 2027 (Expected)
*BSc Electrical and Electronics Engineering* *Istanbul, Türkiye*
*BSc Mathematics, Double Major GPA 3.70/4.00*

**Nanyang Technological University (NTU)** Aug. 2024 – Dec. 2024
*Exchange Program* *Singapore*

**Kayseri Science High School** Sept. 2017 – June 2021
*High school* *Kayseri, Türkiye*

## Experience

**Undergraduate Research** Oct. 2026 – Present
*Technical University of Denmark – DTU Compute* *Denmark*

- Continuing research under the supervision of *Assoc. Prof. Tobias K. S. Ritschel* on stochastic nonlinear Model Predictive Control (MPC) for flexible operation of district heating systems.
- Extended the single-pipe district heating model and reformulated the MPC structure to account for nonlinear thermal dynamics and operational requirements.

*Summer Internship* *July 2026 – Sept. 2026*

- Derived and implemented stochastic nonlinear MPC formulations incorporating chance constraints and uncertainty propagation.
- Integrated Kalman filtering for state estimation and evaluated open-loop and closed-loop controllers through Monte Carlo simulations.

**Undergraduate Research** Jan. 2026 – Present
*Koç University – Robotics and Mechatronics Laboratory (RML)* *Istanbul, Türkiye*

- Continuing research under the supervision of Prof. Çağatay Başdoğan on navigation for the visually impaired via haptic cues.
- Conducted human-subject studies to design and evaluate vibrotactile feedback signals for assistive navigation.
- Developed four haptic navigation strategies for visually impaired individuals using vibrotactile feedback signals identified through human-subject studies.
- Investigating risk-aware MPC for haptic navigation using LSTM-based human motion prediction, conformal prediction for uncertainty estimation, and online adaptation via RLS and LQT.

**Summer Internship** Aug. 2025 – Sept. 2025
*ASELSAN – UGES* *Ankara, Türkiye*

- Worked in the Railway Signalization and Control team within the Department of Transportation Systems Engineering.
- Developed a route decision algorithm and a train simulation model to analyze motor characteristics.

**Research Internship** Aug. 2024 – Dec. 2024
*Nanyang Technological University* *Singapore*

- Contributed to B-Scan imaging of tree trunks using radar for decay detection under the supervision of Prof. Abdulkadir C. Yucel.

**Summer Research Internship** June 2024 – July 2024
*TUBITAK Bilgem* *Istanbul, Türkiye*

- Worked on Differential Cryptanalysis in the Cryptographic Algorithms and Protocol Design Department at the National Research Institute of Electronics and Cryptology (UEKAE).
- Analyzed vulnerability to differential attacks in selected cryptographic algorithms.

**Student Assistant** Oct. 2023 – Jan. 2024
*Bogazici University* *Istanbul, Türkiye*

- *Math 201 - Matrix Theory*

## Communities

**Volunteer Member** Aug. 2024 – Dec. 2024
*Vision of Hope - NTU Welfare Service Club Regular Service Project* *Singapore*

- Supported the visually impaired community through regular programs and events such as house services and outings.

**Director of Robotics & Automation Society (RAS)** June 2022 – June 2023
*BUEC - Bogazici University IEEE Student Branch* *Istanbul, Türkiye*

- Organized a Robot Cup event hosting over 400 participants
- Arranged C++ and Python training sessions for undergraduate students.

## Manuscripts in Preparation

**Stochastic Nonlinear MPC for Flexible Operation of District Heating Systems**
*Taha Berk Koç Supervisor: Tobias K. S. Ritschel* July 2026 – Present
*Target: IEEE CDC 2027*
*Foundational work: DTU Internship Report*

**Risk-Aware Haptic Navigation via Adaptive Conformal MPC for the Visually Impaired**
*Taha Berk Koç, Hilal İlhan Supervisor: Çağatay Başdoğan* Sept. 2026 – Present
*Target: IEEE/RSJ IROS 2027*

**A Vibrotactile Haptic Vest System for Indoor Navigation Assistance for Blind Users**
*Hilal İlhan, Taha Berk Koç Supervisor: Çağatay Başdoğan* Jan. 2026 – Present
*Target: IEEE World Haptics Conference 2027, Boston*

## Selected Projects

**Optimization-Based Approach to Blind Image Deblurring Using Dictionary Learning**
*EE 475 Image Processing Course Project, Boğaziçi University* Fall 2025
*Coauthor: Efe Karakoca*

## Achievements

*National University Entrance Exam* June 2021

- Ranked 394$^{th}$ nationwide among over 2.5 million participants.

*International Mathematical Modeling Challenge (IM2C)* June 2020

- Secured 1$^{st}$ place nationally and received an Honorable Mention internationally.

## Scholarships

***TÜBİTAK 2205*** – *Merit Based Undergraduate Scholarship* Sept. 2023 – Present
***TÜBİTAK 2224-D*** – *Support Program for Student Participation in International Events* 2026

## Relevant Coursework

Linear System Theory
Introduction to Robot Control
System Dynamics and Control
Probability for EE
Numerical Methods for Electrical Engineering
Metric Spaces
Group Theory
Machine Learning Methods and Neural Networks
Nonlinear Dynamics
Control Technology & Design
Signals and Systems I–II
Advanced Calculus I–II
Introduction to Image Processing

## Technical Skills & Interests

**Languages**: Turkish (Native), English (Advanced)
**Programming Languages**: C/C++, MATLAB, Python, Julia, VHDL
**Frameworks & Libraries**: ROS1/2, Gazebo, Isaac Sim, CasADi, SciPy
**Interests**: Music (guitar & violin), Camping (licensed), Archery (licensed), Skiing, Traveling